King Saud University
Muzahimya branch
College of Applied computer science



Subject: Network Protocols

ID Course: cos370

Term: 2026fall

Report title:SMTP

Project No: 2

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Name: Fahad Awad

Network Topology (SMTP & POP3)

Client 1 (A) (SMTP Client)

OS: Windows 7

Hostname: fahad-PC

IP: 189.120.0.57

Port: 49217

User: fahad@mymail.com

Mail Server (B) (hMailServer)

OS: Windows Server 2025

Host: WIN-722GTTLH6IM

IP: 189.120.0.58

SMTP Port: 587

POP3 Port: 110

Domain: mymail.com

Client 2 (A) (POP3 Client)

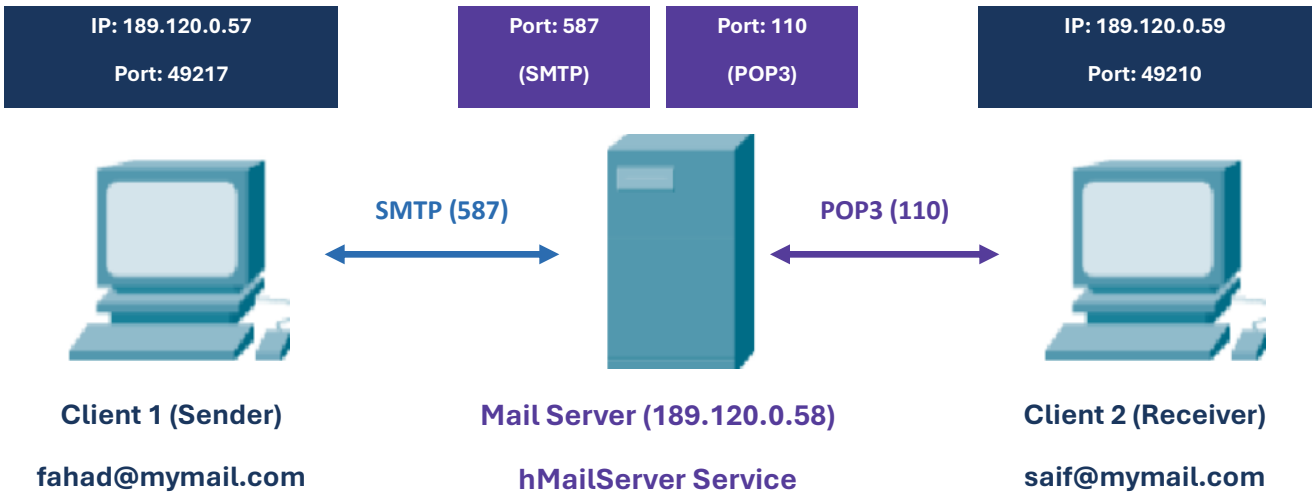
OS: Windows 7

Hostname: saif-PC

IP: 189.120.0.59

Port: 49210

User: saif@mymail.com



Message Table (SMTP)

#	Trace #	IPs	IPd	Ps	Pd	Rcom	Length	Hex
1.	7	189.120.0.58	189.120.0.57	587	49217	220 WIN-722GTTLH6IM ESMTP	27	32 32 30 20 57 49 4e 2d 37 32 32 47 54 54 4c 48 36 49 4d 20 45 53 4d 54 50 0d 0a
2.	8	189.120.0.57	189.120.0.58	49217	587	EHLO [189.120.0.57]	21	45 48 4c 4f 20 5b 31 38 39 2e 31 32 30 2e 30 2e 35 37 5d 0d 0a
3.	9	189.120.0.58	189.120.0.57	587	49217	250-WIN-722GTTLH6IM 250-SIZE 20480000 250-AUTH LOGIN 250 HELP	66	32 35 30 2d 57 49 4e 2d 37 32 32 47 54 54 4c 48 36 49 4d 0d 0a 32 35 30 2d 53 49 5a 45 20 32 30 34 38 30 30 30 30 0d 0a 32 35 30 2d 41 55 54 48 20 4c 4f 47 49 4e 0d 0a 32 35 30 20 48 45 4c 50 0d 0a
4.	10	189.120.0.57	189.120.0.58	49217	587	AUTH LOGIN	12	41 55 54 48 20 4c 4f 47 49 4e 0d 0a
5.	11	189.120.0.58	189.120.0.57	587	49217	334 VXNlcm5hbWU6	18	33 33 34 20 56 58 4e 6c 63 6d 35 68 62 57 55 36 0d 0a
6.	12	189.120.0.57	189.120.0.58	49217	587	ZmFoYWRAbXltYWlsLmNvbQ==	26	5a 6d 46 6f 59 57 52 41 62 58 6c 74 59 57 6c 73 4c 6d 4e 76 62 51 3d 3d 0d 0a
7.	13	189.120.0.58	189.120.0.57	587	49217	334 UGFzc3dvcmQ6	18	33 33 34 20 55 47 46 7a 63 33 64 76 63 6d 51 36 0d 0a
8.	14	189.120.0.57	189.120.0.58	49217	587	MTIzNDU=	10	4d 54 49 7a 4e 44 55 3d 0d 0a
9.	15	189.120.0.58	189.120.0.57	587	49217	235 authenticated.	20	32 33 35 20 61 75 74 68 65 6e 74 69 63 61 74 65 64 2e 0d 0a
10.	16	189.120.0.57	189.120.0.58	49217	587	MAIL FROM:<fahad@mymail.com> SIZE=403	39	4d 41 49 4c 20 46 52 4f 4d 3a 3c 66 61 68 61 64 40 6d 79 6d 61 69 6c 2e 63 6f 6d 3e 20 53 49 5a 45 3d 34 30 33 0d 0a
11.	17	189.120.0.58	189.120.0.57	587	49217	250 OK	8	32 35 30 20 4f 4b 0d 0a
12.	18	189.120.0.57	189.120.0.58	49217	587	RCPT TO:<saif@mymail.com>	27	52 43 50 54 20 54 4f 3a 3c 73 61 69 66 40 6d 79 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a
13.	19	189.120.0.58	189.120.0.57	587	49217	250 OK	8	32 35 30 20 4f 4b 0d 0a
14.	20	189.120.0.57	189.120.0.58	49217	587	DATA	6	44 41 54 41 0d 0a
15.	21	189.120.0.58	189.120.0.57	587	49217	354 OK, send.	15	33 35 34 20 4f 4b 2c 20 73 65 6e 64 2e 0d 0a
16.	22	189.120.0.57	189.120.0.58	49217	587	Message-ID: <8a10fb24-7d56-4b62-8151-7d30555fbac7@mymail.com> Date: Sat, 26 Sep 2026 00:08:58 -0700 MIME-Version: 1.0 User-Agent: Mozilla Thunderbird Content-Language: en-US To: saif@mymail.com From: fahad <fahad@mymail.com> Subject: Test Content-Type: text/plain; charset=UTF-8; format=flowed Content-Transfer-Encoding: 7bit Hello my dear saif i am fahad i am texting you just for fun .	403	4d 65 73 73 61 67 65 2d 49 44 3a 20 3c 38 61 31 30 66 62 32 34 2d 37 64 35 36 2d 34 62 36 32 2d 38 31 35 31 2d 37 64 33 30 35 35 35 66 62 61 63 37 40 6d 79 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 44 61 74 65 3a 20 53 61 74 2c 20 32 36 20 53 65 70 20 32 30 32 36 20 30 30 3a 30 38 3a 35 38 20 2d 30 37 30 30 0d 0a 4d 49 4d 45 2d 56 65 72 73 69 6f 6e 3a 20 31 2e 30 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 20 54 68 75 6e 64 65 72 62 69 72 64 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d 55 53 0d 0a 54 6f 3a 20 73 61 69 66 40 6d 79 6d 61 69 6c 2e 63 6f 6d 0d 0a 46 72 6f 6d 3a 20 66 61 68 61 64 20 3c 66 61 68 61 64 40 6d 79 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 53 75 62 6a 65 63 74 3a 20 54 65 73 74 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f 70 6c 61 69 6e 3b 20 63 68 61 72 73 65 74 3d 55 54 46 2d 38 3b 20 66 6f 72 6d 61 74 3d 66 6c 6f 77 65 64 0d 0a 43 6f 6e 74 65 6e 74 2d 54 72 61 6e 73 66 65 72 2d 45 6e 63 6f 64 69 6e 67 3a 20 37 62 69 74 0d 0a 0d 0a 48 65 6c 6c 6f 20 6d 79 20 64 65 61 72 20 73 61 69 66 20 69 20 61 6d 20 66 61 68 61 64 20 69 20 61 6d 20 74 65 78 74 69 6e 67 20 79 6f 75 20 6a 75 73 74 20 66 6f 72 20 66 75 6e 20 2e 0d 0a 0d 0a
17.	23	189.120.0.57	189.120.0.58	49217	587	.	3	2e 0d 0a
18.	36	189.120.0.58	189.120.0.57	587	49217	250 Queued (0.437 seconds)	28	32 35 30 20 51 75 65 75 65 64 20 28 30 2e 34 33 37 20 73 65 63 6f 6e 64 73 29 0d 0a
19.	42	189.120.0.57	189.120.0.58	49217	587	QUIT	6	51 55 49 54 0d 0a
20.	43	189.120.0.58	189.120.0.57	587	49217	221 goodbye	13	32 32 31 20 67 6f 6f 64 62 79 65 0d 0a

Message Table (POP3)

#	Trace #	IPs	IPd	Ps	Pd	Rcom	Length	Hex
1.	51	189.120.0.58	189.120.0.59	110	49210	+OK POP3	10	2b 4f 4b 20 50 4f 50 33 0d 0a
2.	52	189.120.0.59	189.120.0.58	49210	110	CAPA	6	43 41 50 41 0d 0a
3.	53	189.120.0.58	189.120.0.59	110	49210	+OK CAPA list follows USER UIDL TOP .	43	2b 4f 4b 20 43 41 50 41 20 6c 69 73 74 20 66 6f 6c 6c 6f 77 73 0d 0a 55 53 45 52 0d 0a 55 49 44 4c 0d 0a 54 4f 50 0d 0a 2e 0d 0a
4.	54	189.120.0.59	189.120.0.58	49210	110	USER saif@mymail.com	22	55 53 45 52 20 73 61 69 66 40 6d 79 6d 61 69 6c 2e 63 6f 6d 0d 0a
5.	55	189.120.0.58	189.120.0.59	110	49210	+OK Send your password	24	2b 4f 4b 20 53 65 6e 64 20 79 6f 75 72 20 70 61 73 73 77 6f 72 64 0d 0a
6.	56	189.120.0.59	189.120.0.58	49210	110	PASS 12345	12	50 41 53 53 20 31 32 33 34 35 0d 0a
7.	57	189.120.0.58	189.120.0.59	110	49210	+OK Mailbox locked and ready	30	2b 4f 4b 20 4d 61 69 6c 62 6f 78 20 6c 6f 63 6b 65 64 20 61 6e 64 20 72 65 61 64 79 0d 0a
8.	58	189.120.0.59	189.120.0.58	49210	110	STAT	6	53 54 41 54 0d 0a
9.	59	189.120.0.58	189.120.0.59	110	49210	+OK 2 1028	12	2b 4f 4b 20 32 20 31 30 32 38 0d 0a
10.	60	189.120.0.59	189.120.0.58	49210	110	LIST	6	4c 49 53 54 0d 0a
11.	61	189.120.0.58	189.120.0.59	110	49210	+OK 2 messages (1028 octets)	30	2b 4f 4b 20 32 20 6d 65 73 73 61 67 65 73 20 28 31 30 32 38 20 6f 63 74 65 74 73 29 0d 0a
12.	64	189.120.0.59	189.120.0.58	49210	110	UIDL	6	55 49 44 4c 0d 0a
13.	65	189.120.0.58	189.120.0.59	110	49210	+OK 2 messages (1028 octets) 1 1 2 2 .	43	2b 4f 4b 20 32 20 6d 65 73 73 61 67 65 73 20 28 31 30 32 38 20 6f 63 74 65 74 73 29 0d 0a 31 20 31 0d 0a 32 20 32 0d 0a 2e 0d 0a
14.	66	189.120.0.59	189.120.0.58	49210	110	RETR 2	8	52 45 54 52 20 32 0d 0a
15.	68	189.120.0.58	189.120.0.59	110	49210	+OK 529 octets Return-Path: fahad@mymail.com Received: from [189.120.0.57] (fahad-PC [189.120.0.57]) by WIN-722GTTLH6IM with ESMTPA ; Sat, 26 Sep 2026 10:08:58 +0300 Message-ID: <8a10fb24-7d56-4b62-8151-7d30555fbac7@mymail.com> Date: Sat, 26 Sep 2026 00:08:58 -0700 MIME-Version: 1.0 User-Agent: Mozilla Thunderbird Content-Language: en-US To: saif@mymail.com From: fahad <fahad@mymail.com> Subject: Test Content-Type: text/plain; charset=UTF-8; format=flowed Content-Transfer-Encoding: 7bit Hello my dear saif i am fahad i am texting you just for fun .	576	2b 4f 4b 20 35 32 39 20 6f 63 74 65 74 73 0d 0a 52 65 74 75 72 6e 2d 50 61 74 68 3a 20 66 61 68 61 64 40 6d 79 6d 61 69 6c 2e 63 6f 6d 0d 0a 52 65 63 65 69 76 65 64 3a 20 66 72 6f 6d 20 5b 31 38 39 2e 31 32 30 2e 30 2e 35 37 5d 20 28 66 61 68 61 64 2d 50 43 20 5b 31 38 39 2e 31 32 30 2e 30 2e 35 37 5d 29 0d 0a 09 62 79 20 57 49 4e 2d 37 32 32 47 54 54 4c 48 36 49 4d 20 77 69 74 68 20 45 53 4d 54 50 41 0d 0a 09 3b 20 53 61 74 2c 20 32 36 20 53 65 70 20 32 30 32 36 20 31 30 3a 30 38 3a 35 38 20 2b 30 33 30 30 0d 0a 4d 65 73 73 61 67 65 2d 49 44 3a 20 3c 38 61 31 30 66 62 32 34 2d 37 64 35 36 2d 34 62 36 32 2d 38 31 35 31 2d 37 64 33 30 35 35 35 66 62 61 63 37 40 6d 79 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 44 61 74 65 3a 20 53 61 74 2c 20 32 36 20 53 65 70 20 32 30 32 36 20 30 30 3a 30 38 3a 35 38 20 2d 30 37 30 30 0d 0a 4d 49 4d 45 2d 56 65 72 73 69 6f 6e 3a 20 31 2e 30 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 20 54 68 75 6e 64 65 72 62 69 72 64 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d 55 53 0d 0a 54 6f 3a 20 73 61 69 66 40 6d 79 6d 61 69 6c 2e 63 6f 6d 0d 0a 46 72 6f 6d 3a 20 66 61 68 61 64 20 3c 66 61 68 61 64 40 6d 79 6d 61 69 6c 2e 63 6f 6d 3e 0d 0a 53 75 62 6a 65 63 74 3a 20 54 65 73 74 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f 70 6c 61 69 6e 3b 20 63 68 61 72 73 65 74 3d 55 54 46 2d 38 3b 20 66 6f 72 6d 61 74 3d 66 6c 6f 77 65 64 0d 0a 43 6f 6e 74 65 6e 74 2d 54 72 61 6e 73 66 65 72 2d 45 6e 63 6f 64 69 6e 67 3a 20 37 62 69 74 0d 0a 0d 0a 48 65 6c 6c 6f 20 6d 79 20 64 65 61 72 20 73 61 69 66 20 69 20 61 6d 20 66 61 68 61 64 20 69 20 61 6d 20 74 65 78 74 69 6e 67 20 79 6f 75 20 6a 75 73 74 20 66 6f 72 20 66 75 6e 20 2e 0d 0a 0d 0a
16.	71	189.120.0.59	189.120.0.58	49210	110	QUIT	6	51 55 49 54 0d 0a
17.	72	189.120.0.58	189.120.0.59	110	49210	+OK POP3 server saying goodbye...	35	2b 4f 4b 20 50 4f 50 33 20 73 65 72 76 65 72 20 73 61 79 69 6e 67 20 67 6f 6f 64 62 79 65 2e 2e 2e 0d 0a

IP: 189.120.0.57

Port: 49217 (SMTP)



Client 1

Flow Graph (SMTP)

IP: 189.120.0.58

Port: 587 (SMTP)



Server

220 WIN-722GTTLH6IM ESMTP

EHLO [189.120.0.57]

250-WIN-722GTTLH6IM

250-SIZE 20480000

250-AUTH LOGIN

250 HELP.

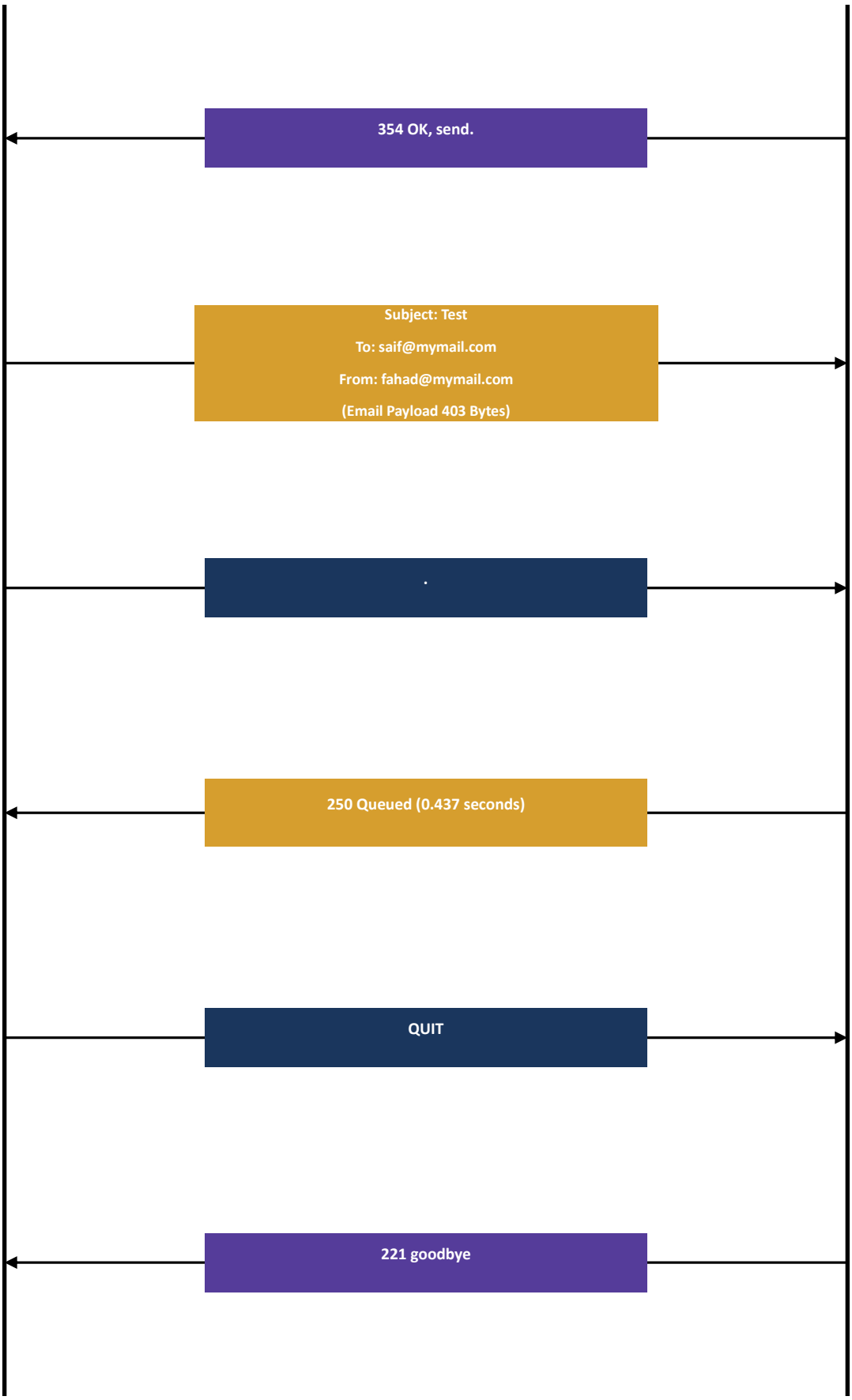
AUTH LOGIN

334 VXNlcm5hbWU6

ZmFoYWRAbXltYWlsLmNvbQ==

334 UGFzc3dvcmQ6


MTIzNDU=
235 authenticated.
MAIL FROM:<fahad@mymail.com> SIZE=403
250 OK
RCPT TO:<saif@mymail.com>
250 OK
DATA


354 OK, send.
Subject: Test
To: saif@mymail.com
From: fahad@mymail.com
(Email Payload 403 Bytes)
.
250 Queued (0.437 seconds)
QUIT
221 goodbye

IP: 189.120.0.59

Port: 49210 (POP3)



Client 2

Flow Graph (POP3)

IP: 189.120.0.58

Port: 110 (POP3)

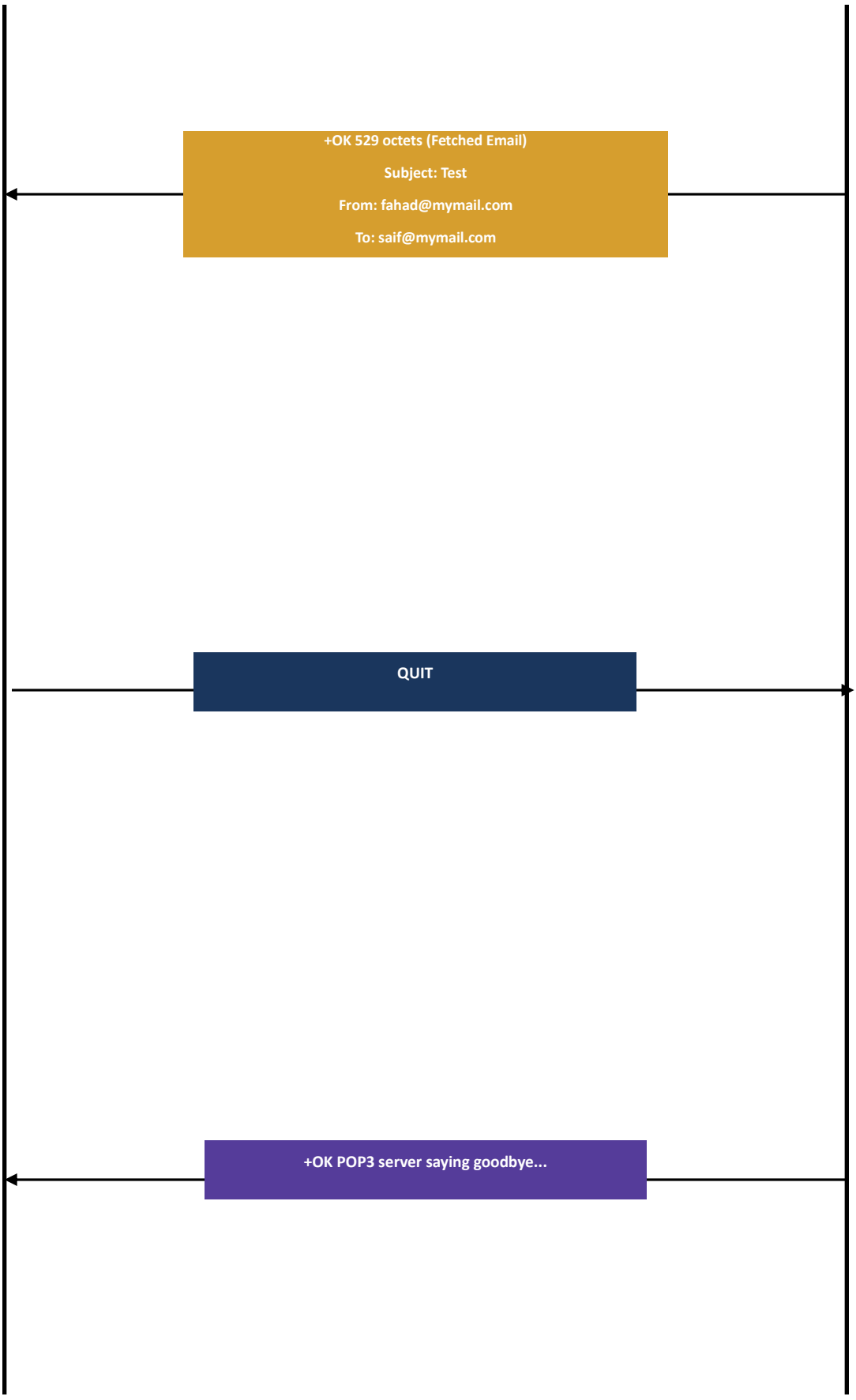


Server




PASS 12345
+OK Mailbox locked and ready
STAT
+OK 2 1028
LIST


+OK 2 messages (1028 octets)
UIDL
+OK 2 messages (1028 octets) 1 1 2 2 .
RETR 2


+OK 529 octets (Fetched Email)
Subject: Test
From: fahad@mymail.com
To: saif@mymail.com
QUIT
+OK POP3 server saying goodbye...

TCP Timeline (SMTP)



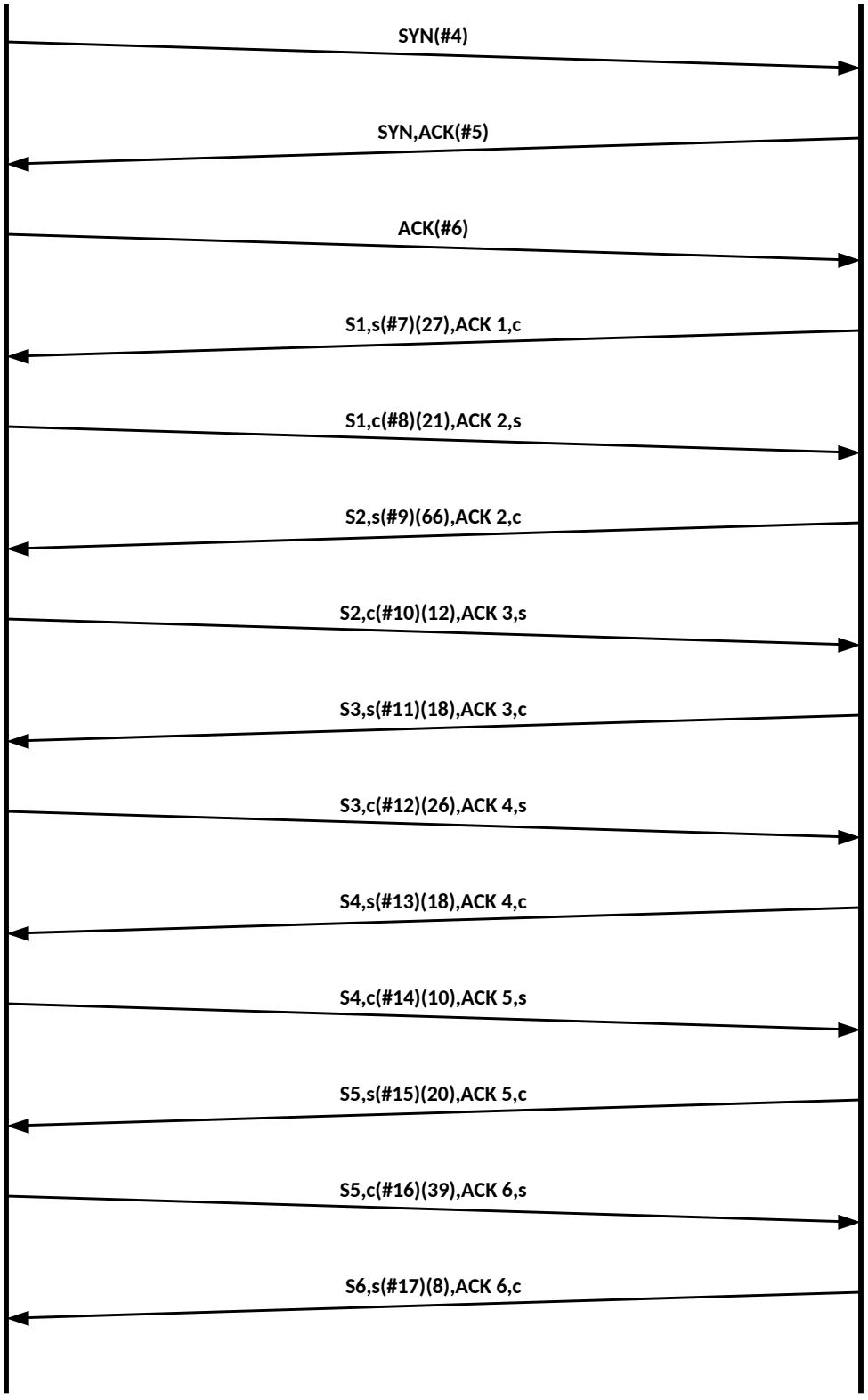
IP: 189.120.0.57
Port: 49217

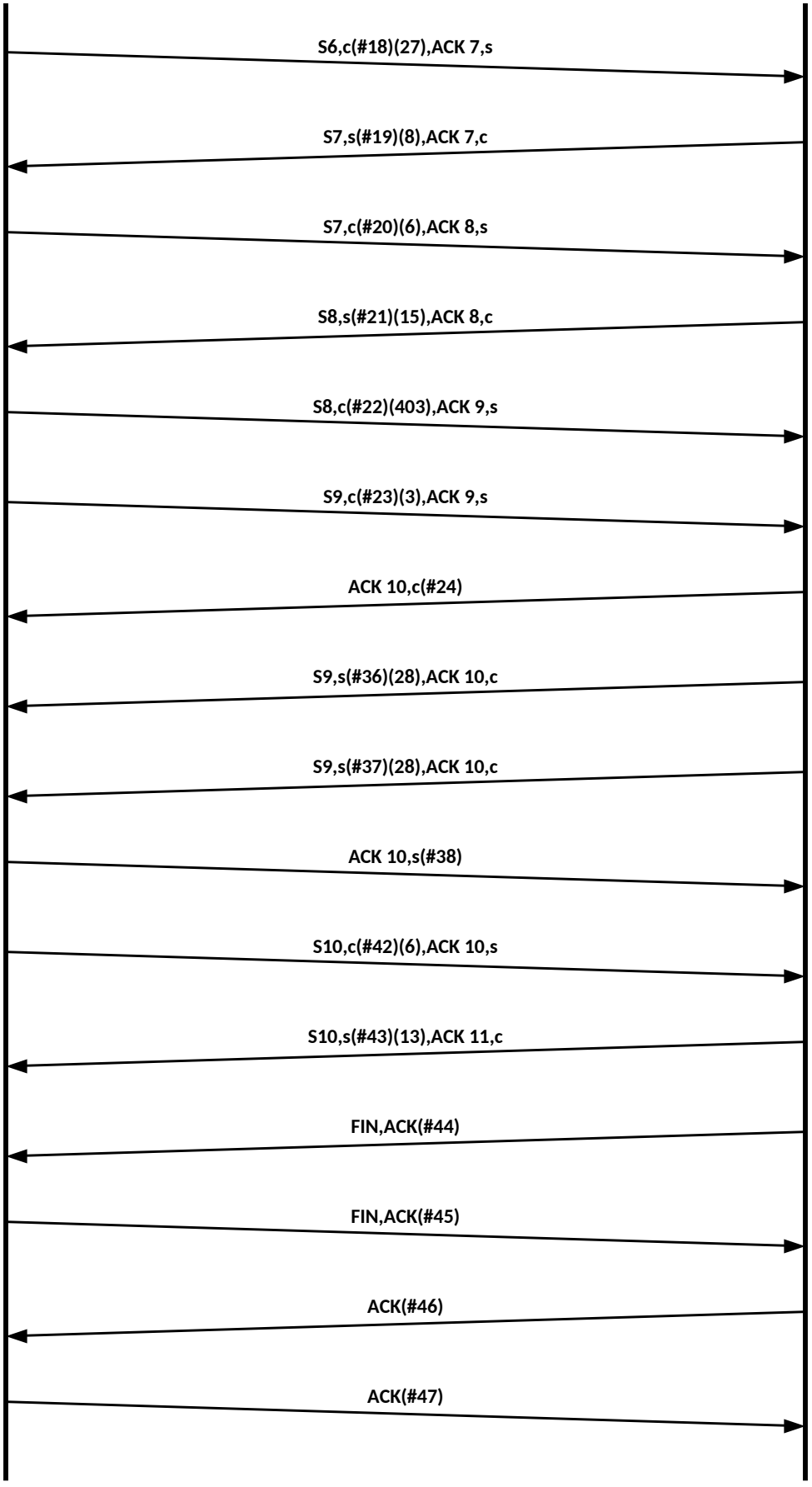
IP: 189.120.0.58
Port: 587



Client 1

Server




S6,c(#18)(27),ACK 7,s
S7,s(#19)(8),ACK 7,c
S7,c(#20)(6),ACK 8,s
S8,s(#21)(15),ACK 8,c
S8,c(#22)(403),ACK 9,s
S9,c(#23)(3),ACK 9,s
ACK 10,c(#24)
S9,s(#36)(28),ACK 10,c
S9,s(#37)(28),ACK 10,c
ACK 10,s(#38)
S10,c(#42)(6),ACK 10,s
S10,s(#43)(13),ACK 11,c
FIN,ACK(#44)
FIN,ACK(#45)
ACK(#46)
ACK(#47)

TCP Timeline (POP3)



SYN(#48)

SYN,ACK(#49)

ACK(#50)

S1,s(#51)(10),ACK 1,c

S1,c(#52)(6),ACK 2,s

S2,s(#53)(43),ACK 2,c

S2,c(#54)(22),ACK 3,s

S3,s(#55)(24),ACK 3,c

S3,c(#56)(12),ACK 4,s

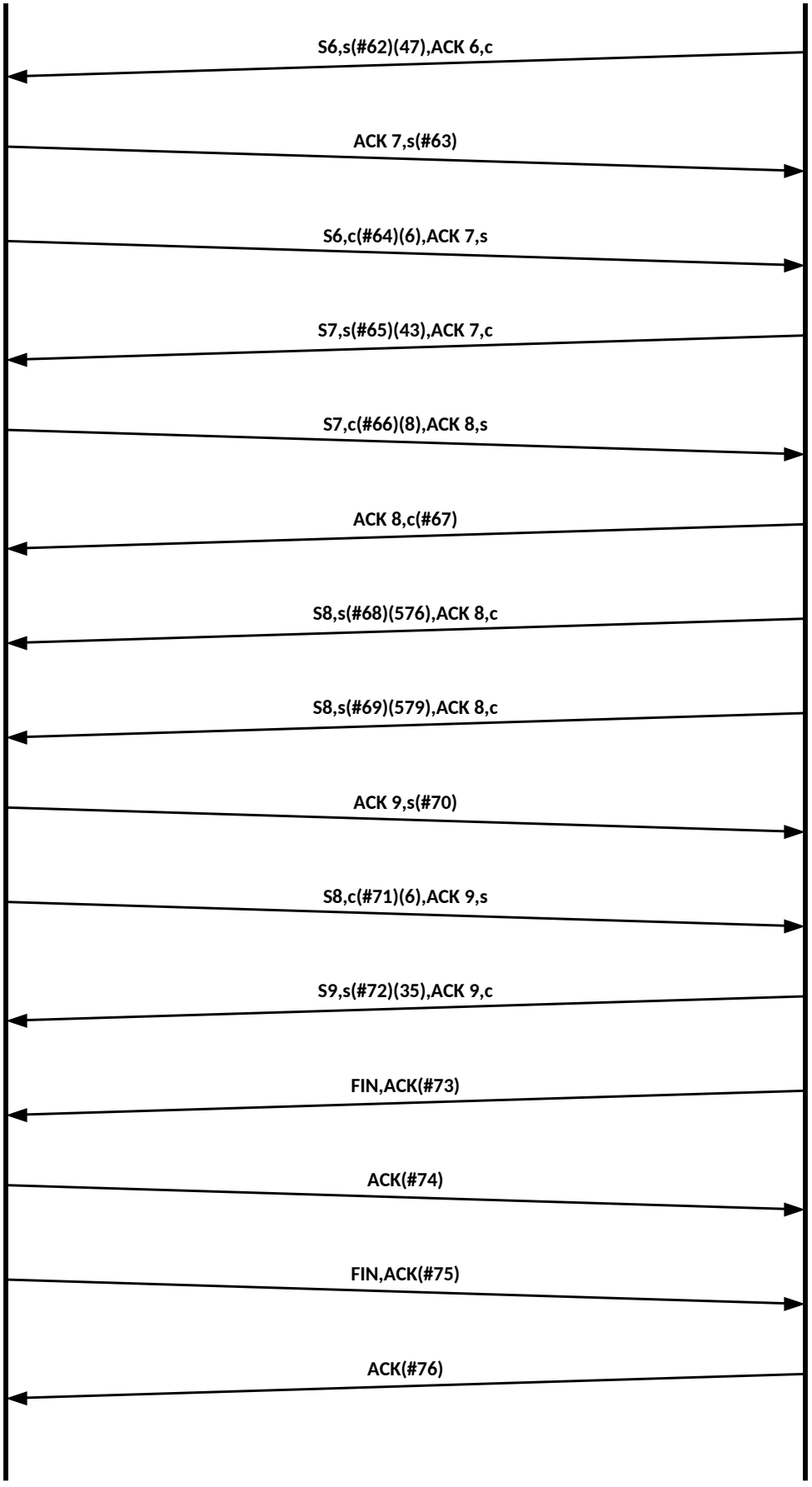
S4,s(#57)(30),ACK 4,c

S4,c(#58)(6),ACK 5,s

S5,s(#59)(12),ACK 5,c

S5,c(#60)(6),ACK 6,s

S6,s(#61)(30),ACK 6,c


S6,s(#62)(47),ACK 6,c
ACK 7,s(#63)
S6,c(#64)(6),ACK 7,s
S7,s(#65)(43),ACK 7,c
S7,c(#66)(8),ACK 8,s
ACK 8,c(#67)
S8,s(#68)(576),ACK 8,c
S8,s(#69)(579),ACK 8,c
ACK 9,s(#70)
S8,c(#71)(6),ACK 9,s
S9,s(#72)(35),ACK 9,c
FIN,ACK(#73)
ACK(#74)
FIN,ACK(#75)
ACK(#76)